# Marriage and Divorce Dynamics in Denmark

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# Plan for today

- Bruze, Svarer and Weiss (2015): "The Dynamics of Marriage and Divorce"
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- **Reading guide:**
  1. What are the main *research questions*?
  2. What is the *(empirical) motivation*?
  3. What are the central *mechanisms in the model*?
  4. What is the *simplest model* in which we could capture these?

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- **Reading guide:**
  1. What are the main *research questions*?
     - How does marriage and divorce behavior vary across age and educational groups?
     - How does educational differences influence intra-household inequality?
  2. What is the *(empirical) motivation*?
  3. What are the central *mechanisms in the model*?
  4. What is the *simplest model* in which we could capture these?

# Empirical Motivation: I

- **Marriage and divorce**
  Age of female cohort = age of male cohort-2

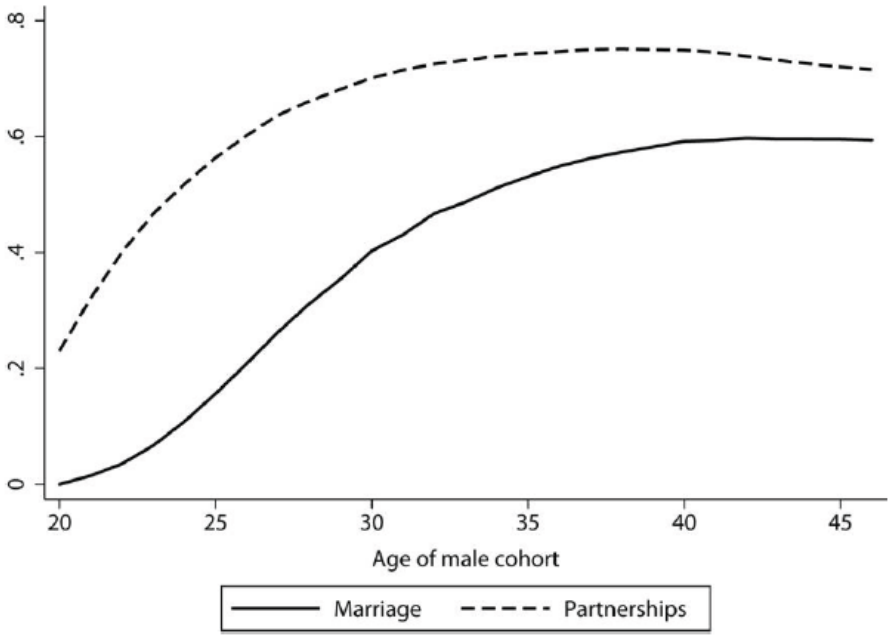


FIG. 1.—Fraction married or in partnerships (marriage plus cohabitation) by age. A color version of this figure is available online.

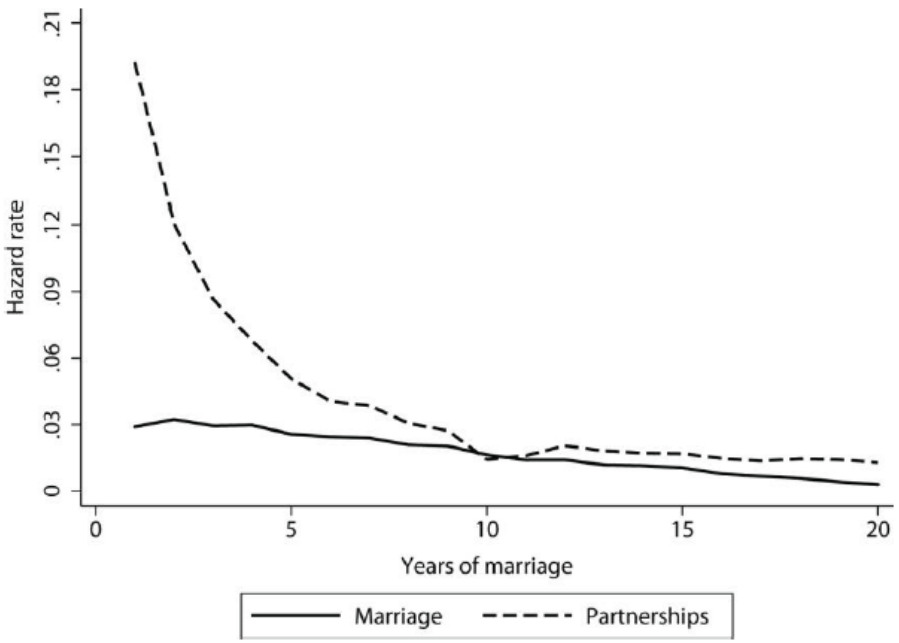


FIG. 3.—Divorce hazard for first marriage or partnership (marriage plus cohabitation). A color version of this figure is available online.

# Empirical Motivation: II

- **Highly educated** people partner later but more "stable". Especially if both highly educated

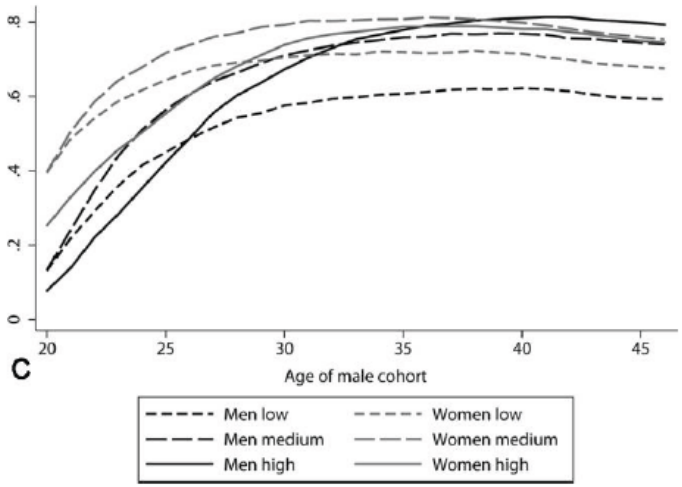


FIG. 4.—*A*, Fraction married men and women by age and education; *B*, fraction cohabiting men and women by age and education; *C*, fraction men and women in partnerships by age and education. A color version of this figure is available online.

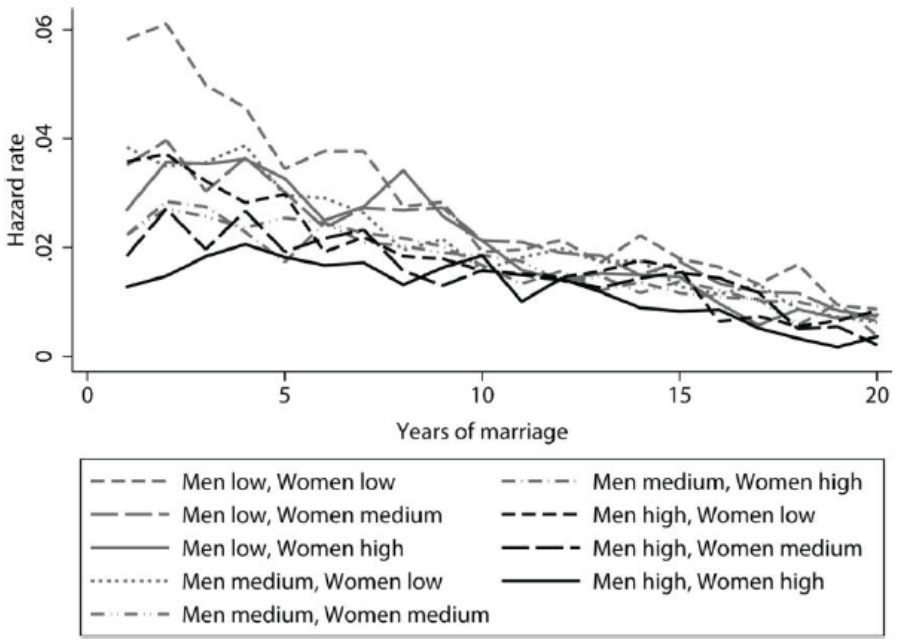


FIG. 6.—Divorce hazards for first marriages by education of the husband and wife. A color version of this figure is available online.

# Empirical Motivation: III

- **Highly educated** people re-marry faster.
  And stay in second marriage longer.

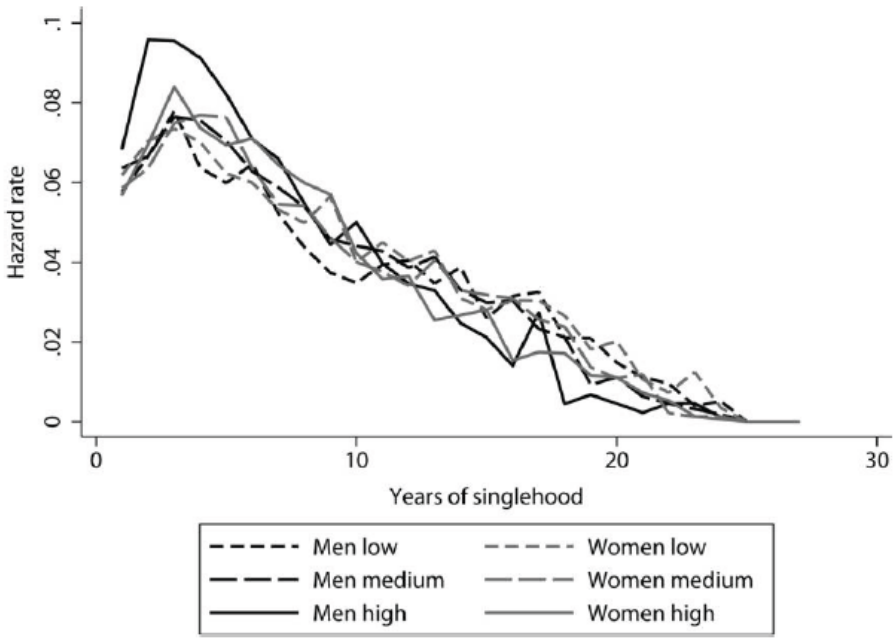


FIG. 7.—Hazard rate into second marriage for men and women by education. A color version of this figure is available online.

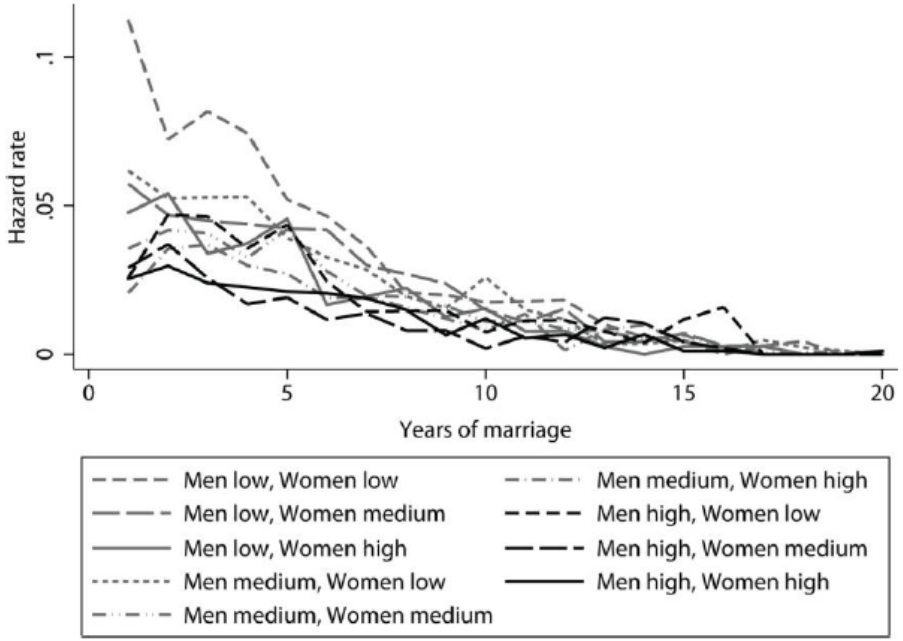


FIG. 8.—Divorce hazards when at least one spouse is in second marriage, by education of the husband and wife. A color version of this figure is available online.

# Empirical Motivation: IV

- **Assortative matching:**
  people more likely to marry one with same education

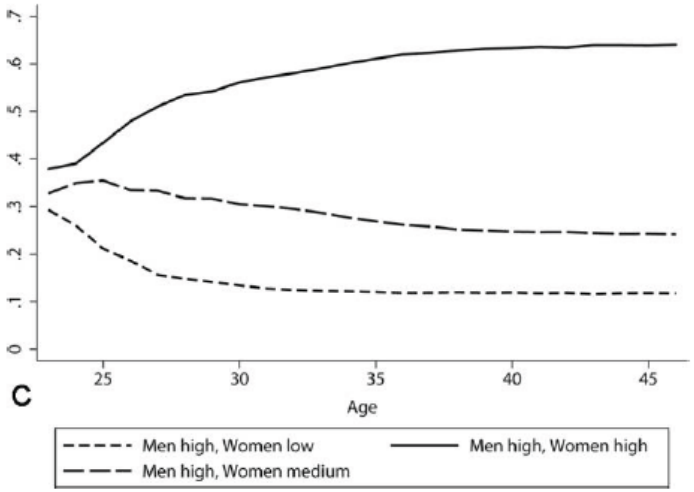


FIG. 9.—Distribution of marriages for men with low (top), medium (middle), or high (bottom) education. A color version of this figure is available online.

(a) $P(educ_w | educ_m = high)$

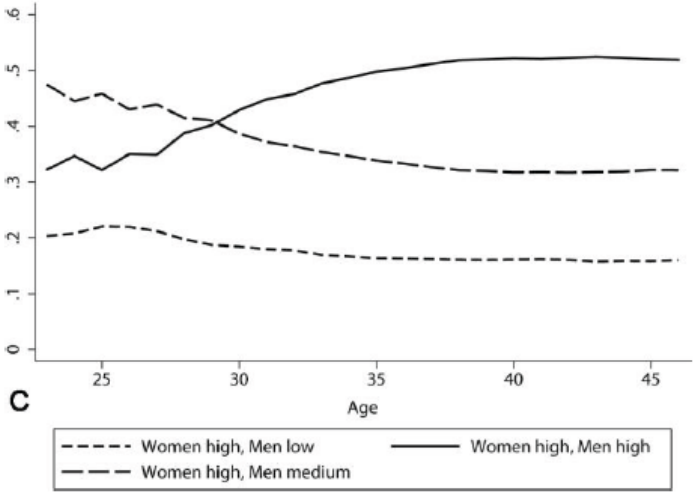


FIG. 10.—Distribution of marriages for women with low (top), medium (middle), or high (bottom) education. A color version of this figure is available online.

(b) $P(educ_m | educ_w = high)$

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  - Structure $\rightarrow$ identify bargaining power $\rightarrow$ intra-household inequality
    Fundamentally unobserved
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    Fundamentally unobserved
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- **From Abstract:**
  *Education raises the share of the marital surplus for men but not for women. As men and women get older, husbands receive a larger share of the marital surplus*

# Outline

1 Model and Mechanisms

2 Estimation

# Model Overview

- **Full commitment:**
  Transferable utility
  Perfect foresight wrt bargaining power.
  Particular timing/expectation assumptions (get back)
- **Choices:**
  Marriage: which type $IJ$
  Divorce
- **States:**
  $d_t$: duration of marriage
  "TYPES":
  $e \in E = \{l, m, h\}$: Educational type of both members
  $p_t \in P = \{nm, pm\}$: never/previously married
  $u \in U = \{1, 2\}$ (unobserved type)
  $\rightarrow I \in E \times P \times U$ (men) and $J \in E \times P \times U$ (women)
  (love-shock, $\theta_t \sim iid\mathcal{N}(0, 1)$)

# Bellman Equation: Married

- **Bellman equation** for type $IJ$ (remaining) couple is

$$\underbrace{W_t^{IJ}(d_t)+\theta_t}_{V_t^{m\to m}} = \underbrace{\zeta^{IJ}+\theta_t}_{U^{IJ}} + R\mathbb{E}_t\underbrace{[\max\{\underbrace{W_{t+1}^{IJ}(d_{t+1})+\theta_{t+1}}_{V_{t+1}^{m\to m}}, \underbrace{V_{t+1}^{I}+V_{t+1}^{J}-s(d_{t+1})}_{V_{t+1}^{m\to s}}\}]}_{V_{t+1}^{m}}$$

where
$\zeta^{IJ}$: type-specific utility
$R$: discount factor
$s(d_{t+1})$: divorce cost
$V_{t+1}^{I}+V_{t+1}^{J}$: sum of value of singlehood (TU)
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- **Probability** of observing divorce, $P_D(I,J,t,d_t)$:

$$\Pr(W_t^{IJ}(d_t)+\theta_t < V_t^{I}+V_t^{J}-s(d_t)) = \Phi(V_t^{I}+V_t^{J}-s(d_t)-W_t^{IJ}(d_t))$$

# Bellman Equation: Single

- **Bellman equation** for type $I$ man being single is

$$V_t^I = \varphi^I + R\mathbb{E}_t[V_{t+1}^I + \max_{J \in E \times P \times U}\{\varepsilon_{t+1}^0, \gamma_{t+1}^{IJ}[W_{t+1}(\underbrace{1}_{d_{t+1}}) - V_{t+1}^I - V_{t+1}^J] + \varepsilon_{t+1}^J\}]$$

where
$\varepsilon_{t+1}^0$: EV taste-shock wrt value of singlehood
$\varepsilon_{t+1}^J$: EV taste-shock wrt value of marriage with type $J$
$\gamma_{t+1}^{IJ}$: share of (new) marital surplus to man. Focus in a bit.
$\mathbb{E}_t[]$ is wrt. Extreme Value taste shocks over type of female match, $J$.
(See discussion on following slides.)

- **Value of marriage next period** is thus the value of being single + the share of the marital surplus he gets.
- **Symmetric** for women with share $1 - \gamma_{t+1}^{IJ}$.

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- **Marital surplus** is "special" since future love-shock does not enter...:

$$W_{t+1}(1) - V_{t+1}^I - V_{t+1}^J = \mathbb{E}_t[W_{t+1}(1) + \theta_{t+1} - V_{t+1}^I - V_{t+1}^J]$$

since $\mathbb{E}_t[\theta_{t+1}] = 0$.

# Bellman Equation: Single, Marital Surplus!

- They interpret this as a timing-thing (p. 140):
  *"The quality of match ... is revealed to the partners only at the end of each period. ...*
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- **"Wrong":** This means that the expected value is inserted in the max, rather than taking the expected value of the max...:

$$\max_{J \in E \times P \times U} \{\varepsilon^0_{t+1}, \gamma^{IJ}_{t+1} \mathbb{E}_t [W_{t+1}(1) + \theta_{t+1} - V^I_{t+1} - V^J_{t+1}] + \varepsilon^J_{t+1}\}$$

  vs "correct"

$$\mathbb{E}_t [\max_{J \in E \times P \times U} \{\varepsilon^0_{t+1}, \gamma^{IJ}_{t+1} [W_{t+1}(1) + \theta_{t+1} - V^I_{t+1} - V^J_{t+1}] + \varepsilon^J_{t+1}\}]$$

- **Their formulation removes a numerical integral wrt. $\theta_{t+1}$**
  $\rightarrow$ Speeds up the solution...

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- The expectation in

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- **Known in closed-form:** The log-sum!

$$\mathbb{E}_t[V_{t+1}^I + \max_{J \in E \times P \times U}\{\varepsilon_{t+1}^0, \gamma_{t+1}^{IJ}[W_{t+1}(1) - V_{t+1}^I - V_{t+1}^J] + \varepsilon_{t+1}^J\}]$$

$$=$$

$$\log\left[\exp(V_{t+1}^I) + \sum_{J \in E \times P \times U} \exp(V_{t+1}^I + \gamma_{t+1}^{IJ}[W_{t+1}(1) - V_{t+1}^I - V_{t+1}^J])\right]$$

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- **Probability** (logit) of entering marriage with type $j$, ($j = 0 \rightarrow$single)

$$P_M^I(j,t) = \frac{\exp(V_t^I + \gamma_t^{Ij}[W_t(1) - V_t^I - V_t^j])}{\exp(V_t^I) + \sum_{J \in E \times P \times U} \exp(V_t^I + \gamma_t^{IJ}[W_t(1) - V_t^I - V_t^J])}$$

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- **Assume** the functional form

$$\gamma_t^{IJ} = \frac{\exp\{\rho^{IJ} + \kappa^{IJ} t + \lambda^{IJ} t^2\}}{1 + \exp\{\rho^{IJ} + \kappa^{IJ} t + \lambda^{IJ} t^2\}}$$

which has 108 estimated parameters.
(not reported)

# Outline

1 Model and Mechanisms

2 Estimation

# Remaining Parameters

- **Cost of divorce** "non-parametric" (10)

$$s(d_t) = \sum_{k=1}^{9} \beta_k \mathbf{1}(d_t = k) + \beta_{10} \mathbf{1}(d_t \geq 10)$$

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- **Utility** for singles are $u \in \{1, 2\}$

$$\varphi_t^I = \mu_t^I + \eta_u^I$$
$$\varphi_t^J = \mu_t^J + \eta_u^J$$

and estimated parameters are

$$\zeta^{IJ} \text{ :13 (education mix (9) or marital order mix (4))}$$
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- **Unobserved types** (4):
estimate $u_2^I, u_2^J$ (relative to type 1) and the share of type 2.

# Estimation

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- The (conditional) likelihood function of the observed data is

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assuming independence.

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  assuming independence.
- The EV assumption makes $\Pr(\bullet)$ conditional multinomial logit (MNL) Can be found in closed form.

# Estimation

- **Likelihood of *sequence*** of choices given $S_{i,0}$, $u_i$

$$\Pr(O_i|S_{i,0}, u_i) = \prod_{t=2}^{T} \Pr(O_{i,t}|O_{i,t-1}, u_i) \Pr(O_{i,1}|S_{i,0}, u_i)$$

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- **Do not observe** $u$; we "integrate that out":

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- **The likelihood** of observing the outcomes is then

$$\begin{aligned}L = &\prod_{i=1}^{N^m} [q^m \Pr(O_i|S_{i,0}, u_i = 1) + (1 - q^m) \Pr(O_i|S_{i,0}, u = 2)] \\ &\times \prod_{j=1}^{N^f} [q^f \Pr(O_j|S_{j,0}, u_j = 1) + (1 - q^f) \Pr(O_j|S_{j,0}, u_j = 1)]\end{aligned}$$

# Identification (idea)

- **Identification** arguments in paper
  Only without unobserved types

- **Talk about some here**
  To give idea of arguments
  Ignores unobserved types, $u \in \{1, 2\}$
  $\rightarrow I, J \in E \times P$ (educ and pre-marital status)

# Identification: Weights

- **From probability of marriage** of $I$ with $J$, relative to remaining single

$$\log\left(\frac{P_M^I(J,t)}{P_M^I(0,t)}\right) = \gamma_t^{IJ}[W_t^{IJ}(1) - V_t^I - V_t^J]$$

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- such that taking ratios **identifies weights**, $\gamma_t^{IJ}$:

$$\frac{\gamma_t^{IJ}}{1-\gamma_t^{IJ}} = \underbrace{\log\left(\frac{P_M^I(J,t)}{P_M^I(0,t)}\right) / \log\left(\frac{P_M^J(I,t)}{P_M^J(0,t)}\right)}_{\text{data}}$$

# Identification: Divorce costs

- **From probability of divorce**:

$$V_t^I + V_t^J - s(d_t) - W_t^{IJ}(d_t) = \underbrace{\Phi^{-1}(P_D(I, J, t, d_t))}_{\text{data}}$$

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&\Updownarrow \\
s(1) &= \underbrace{\log\left(\frac{P_M^I(J,t)}{P_M^I(0,t)}\right)}_{\text{data}} / \underbrace{\gamma_t^{IJ}}_{\text{"known"}} + \underbrace{\Phi^{-1}(P_D(I, J, t, 1))}_{\text{data}}
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\end{aligned}$$

- Remaining $s(d)$: Noting that $W_t^{IJ}(d_t)$ depends on $d_t$ through $s(d)$ and

$$\underbrace{\Phi^{-1}(P_D(I, J, t, d_t)) - \Phi^{-1}(P_D(I, J, t, d_t'))}_{\text{data}} = s(d_t') - s(d_t) + W_t^{IJ}(d_t') - W_t^{IJ}(d_t)$$

# Identification: Utility Flow

- **Assume** that flow-utility in couple are constant
- **"Normalize"** value of singlehood for men over age 40 to zero (but more than normalization since several periods, $T = 71$?)
- **"Normalize"** value of singlehood for women over age 38 to zero (but more than normalization since several periods, $T = 71$?)

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- **Couples, surplus:** For $t > 39$:

$$W_t^{IJ}(d_t) + \theta_t - (V_t^I + V_t^J - s(d_t)) = W_t^{IJ}(d_t) + \theta_t + s(d_t)$$

and thus gets $\zeta^{IJ}$ from

$$P_D(I, J, t, d_t) = \Pr(\theta_t \leq -W_t^{IJ}(d_t) - s(d_t)) = \Phi(-W_t^{IJ}(d_t) - s(d_t))$$

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- **Singles:** time-differences in likelihood gives $\varphi_t^I$ and $\varphi_t^J$.

# Results: Marriage Order

- **Estimates** suggest that second marriages are less "costly" for men

Table 3
Effects of Marriage Order on the Marital Output Flow

| | Wife's First Marriage | Wife's Second Marriage |
|---|---|---|
| Husband's first marriage | .5166 | .3891 |
| Husband's second marriage | .4709 | .5364 |

# Results: Divorce Costs

- **Estimates** suggest that divorce costs are U-shaped
  Authors are surprised, but this could still be due to children.

Table 4
Costs of Divorce by Duration of Marriage

| Marital Duration | Cost of Divorce |
|---|---|
| 1 year | 14.3 |
| 2 years | 14.1 |
| 3 years | 12.4 |
| 4 years | 11.5 |
| 5 years | 11.6 |
| 6 years | 11.6 |
| 7 years | 11.5 |
| 8 years | 12.7 |
| 9 years | 12.7 |

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is the *expected value of being forced to remain single*.

- **If allowed on the marriage market**, the expected value is (log-sum):

$$C_{t+1}^I = \mathbb{E}_t[V_{t+1}^I + \max_{J \in E \times P \times U} \{\varepsilon_{t+1}^0, \gamma_{t+1}^{IJ}[W_{t+1}(1) - V_{t+1}^I - V_{t+1}^J] + \varepsilon_{t+1}^J\}$$

- **Expected gains** from entering the marriage market:

$$S_t^I = C_{t+1}^I - \mathbb{E}_t[V_{t+1}^I + \varepsilon_{t+1}^0]$$

and *similarly for women.*

# Results: Marital Surplus Shares

- They define the total share of surplus of the husband be

$$\Gamma_t^{IJ} = \frac{S_t^I}{S_t^I + S_t^J}$$

Table 9
Estimated Average Total Surplus Share Γ for Husband by Education of Husband and Wife

| | Wife's Education | | |
|---|---|---|---|
| Husband's Education | Low | Medium | High |
| Low | .417 | .387 | .402 |
| Medium | .496 | .463 | .490 |
| High | .530 | .493 | .498 |

# Results: Marital Surplus Shares

Table 11
Estimated Average Total Surplus Share Γ for Husband by Marital History of Husband and Wife

| | Wife's First Marriage | Wife's Second Marriage |
|---|---|---|
| Husband's first marriage | .464 | .353 |
| Husband's second marriage | .563 | .455 |

Table 12
Estimated Average Total Surplus Share Γ for Husband by Age of Husband

| Age of Husband | Share of Gains to Marriage |
|---|---|
| 25 | .425 |
| 30 | .465 |
| 35 | .486 |
| 40 | .505 |
| 45 | .541 |

# Next Time

- **Next time:**
  Fertility and Labor Supply.

- **Literature:**
  Jakobsen, Jørgensen and Low (2025): "Fertility and Family Labor Supply"
  [unitary]
  - **Read** before lecture
  - **Reading guide:**
    Section 1: Introduction – Read.
    Section 2: Data. Skim.
    Section 3: Empirical motivation. Get idea.
    Section 4: Model. Key. Get the idea.
    Section 5: Estimation results. skim/read.
    Section 6: Simulation results. Key - read.
    Section 7: Sensitivity/robustness. Can drop.

# References I

Bruze, G., M. Svarer and Y. Weiss (2015): "The Dynamics of Marriage and Divorce," *Journal of Labor Economics*, 33(1), 123–170.

Jakobsen, K., T. H. Jørgensen and H. Low (2025): "Fertility and Family Labor Supply," Working paper, Centre for Economic Behavior and Inequality.